CityFM: from OpenStreetMap to embeddings — the Leeds case study

Method of Balsebre et al. (CIKM 2024) as implemented in this repo and run for the Leeds district. Current model: **v5** (Leeds_v5/Model/model.pkl), which reads every entity by its type tags, without names.

T text embedding (768-d) · V visual embedding (768-d) · P position (256-d) · ⟷ more similar · - - → less similar (decoys) · trained · frozen · **Leeds** = measured for Leeds · A data product (column 2)

1 Input: OpenStreetMap

Overpass download of all nodes, ways and relations inside the Leeds boundary

POIs nodes

amenity, shop, tourism, leisure, office, healthcare or sport tag; benches, bins, bike racks dropped

Leeds: 9,429

feeds A entity text

Buildings tagged polygons

building=* except yes, roof, garage, terrace, shed; or an amenity / shop / office / healthcare tag. building=yes is kept aside for downstream use

Leeds: 262,729 (88% are houses)

feeds A text D footprint

Road segments open ways

highway=* except footway, cycleway and steps; 1 segment = 1 OSM way

Leeds: 55,439 (38% service roads)

feeds B context

Bus routes relations

route=bus relations with at least 7 road-segment members

Leeds: 153 routes

feeds C bus weight

Not used

land-use, park, water and rail shapes; footways and cycleways; all other relations; every tag outside the nine kept in A

2 Pre-processing

CityFM_preprocess.py → Leeds/Data/*.json, then CityFM_collapse.py → Leeds_v2/Data

A Entity text

Each POI or building becomes one string of its type tags (amenity, shop, tourism, healthcare, office, leisure, sport, building). Names are left out.

`shop: bakery`

`building: house`

No names or coordinates, so every POI of the same type gets the same embedding

B Road-segment context



Every entity within 100 m of any vertex of the segment (a set, unordered)

Unnamed house-type buildings are merged into one entry: 'building: few / some / many / very many houses'

Leeds: 8.8 entries per segment (3.4 POIs, 4.6 other buildings, 0.7 house entries)

C Bus-route weight per segment

l = number of bus routes using the segment ÷ the largest such number

Leeds: largest = 61 routes; only 10% of segments have l > 0

D Footprint image + area



Rotated and scaled to fill the image, so it carries shape only; area enters as min(m², 5,000) ÷ 5,000. Made on the fly in stage 2.

Drawn in metres (longitude × cos latitude), so shapes keep their true proportions

3 Pre-training: three contrastive stages

CityFM_train.py --no-names runs the stages in order, each continuing from the previous weights, and saves Leeds_v5/Model/model.pkl. The model holds two encoders, T and V. Each stage trains one of them.

model.pkl · the trained model, with two encoders

T text encoder
tag string (no names) → BERT-base → mean over tokens (and over entities, for a context) → 2-layer MLP → 768-d

V visual encoder
image → ResNet-18 → linear → 768-d; area → small MLP → 768-d; V = mean of the two
ResNet-18 from ImageNet weights

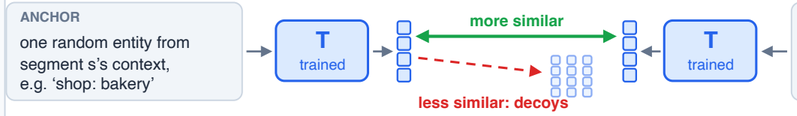
not in the model · a fixed formula

P position code
lat, lon → fixed sin/cos waves at 64 frequencies → 256-d. Not learned. Joined to T and V only for tasks.

Every stage is a matching quiz on unlabelled data. A batch holds N true (anchor, positive) pairs. The InfoNCE loss (temperature 0.5) is low when each anchor is more similar to its own positive than to the other N – 1 positives and the other N – 1 anchors, which act as decoys. The same holds for each positive.

Stage 1 · Infomax entity ↔ the context it sits in

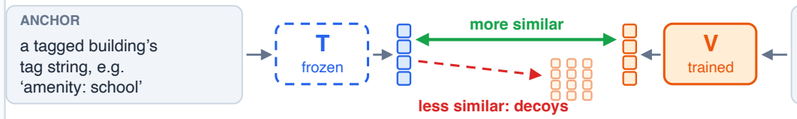
paper: text-based objective · uses B



Trains T (BERT + MLP), lr 1e-4 decaying to 0. Quiz: 64 segments; decoys are the other 63 contexts and 63 entities. 625 steps.
Learns: entities found along the same streets get similar embeddings (e.g. pub moves towards theatre, hostel).
Leeds v5: with the POI held out, it finds the right street 7.0% of the time (untuned BERT 4.2%, guessing 1.6%); categories found together become similar (ρ 0.62, against 0.18 for untuned BERT).

Stage 2 · Polygon footprint ↔ its own tags

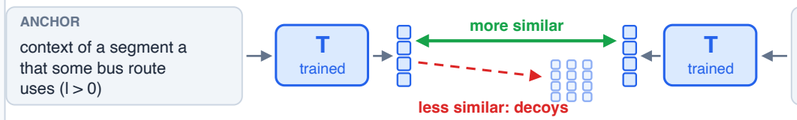
paper: vision–language objective · uses A D



Trains V only (ResNet-18 + projections), lr 1e-3; T is frozen. Quiz: 64 buildings, distinct tags. 1,250 steps.
Learns: a building's V becomes most similar to the T of its own type.
Leeds v5: a building's class (of 18) can be read from its footprint 57% of the time (floor area alone 19%); matching footprints straight to class texts works 47% of the time (chance 6%).

Stage 3 · Relation street ↔ street with similar bus use

paper: road-based objective · uses B C



Trains T (BERT + MLP), lr 1e-5 decaying to 0. Pairs from windows of 1,024 segments: ≈ 10 per quiz. 270 steps.
Learns: segments with a similar role in the bus network get similar context embeddings.
Leeds v5: barely learns (the loss ends 4% of the way from guessing to the floor); only the 10% of segments on a bus route take part.

4 Embeddings

encoder.py on the frozen model.pkl (no names). V can be compared with T. P is not learned.

	T text	V visual	P position
POI a node	its tags	–	its point
Building a tagged polygon	its tags	footprint	centroid
Road segment not in encoder.py	its context	–	a vertex
Area e.g. an Output Area	all inside	buildings	centroid

frozen embeddings + labels

5 Downstream tasks

CityFM_downstream.py: embeddings stay frozen; a small MLP is trained per task



pop_density
area embedding → people per km²
Leeds: Census 2021, 500 m grid + Output Areas

build_func
untagged building (footprint + context + position) → land-use class

avg_speed
road segment → mean traffic speed
shipped code never calls the model

Leeds (500 m grid, R²): v5 features 0.74; 9 simple OSM counts 0.76; counts + position 0.80, so CityFM adds nothing to the counts. Next: IMD, EPC built form / age, DfT traffic counts.

Where this code differs from the paper

- The paper sums the three losses into one; the code trains the three stages one after another.
- Stages 1 and 2 stop each epoch after a set number of items (-m), however big the city; v5 used 32,000 and 64,000.
- Stage 3 pairs segments whose weights differ by ≤ 0.02 (paper: θ = 0.05).
- Added for Leeds: houses merged in contexts; ImageNet weights; footprints in metres; learning-rate fix; no names.

Leeds case study

- Current model v5: trained from scratch on the research server (CPU, ~12 h) → Leeds_v5/Model/model.pkl.
- Inputs: Leeds_v2/Data (houses merged); names left out of all text (--no-names), in training and when embedding.
- Stages 1 and 2 learn; stage 3 barely does. Planned next: replace its bus-route target with street-network metrics.
- The Leeds figures here come from notebooks/leeds_embeddings.ipynb (sections 6–10).